Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/29/2005
PI Name:	Young, Gerald W Ph.D.		
Project Title:	Theory, Modeling, Software and Hardware Development in Computational Materials Science		
Division Name:	Physical Sciences		
Program/Discipline:	OTHER		
Program/Discipline--Element/Subdiscipline:	OTHER--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gwyoung@uakron.edu	**Fax:**	FY 330-374-8630
PI Organization Type:	UNIVERSITY	**Phone:**	330-972-5731
Organization Name:	University of Akron		
PI Address 1:	Department of Theoretical and Applied Mathematics		
PI Address 2:			
PI Web Page:			
City:	Akron	**State:**	OH
Zip Code:	44325-4002	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/07/2004	**End Date:**	03/03/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chait, Arnon	**Contact Phone:**	216-433-3558
Contact Email:	Arnon.Chait@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB27G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The focus of this Cooperative Agreement between the Computational Materials Laboratory (CML) of the Processing Science and Technology Branch of the NASA Glenn Research Center (GRC) and the Department of Theoretical and Applied Mathematics at The University of Akron is in the areas of system development of the CML workstation environment, modeling of microgravity and earth-based material processing systems, and joint activities in laboratory projects. These efforts complement each other as the majority of the modeling work involves numerical computations to support laboratory investigations.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	The primary goal was continued integration of the Computational Materials Laboratory into a smoothly working unit capable of timely response to different needs. Hence, the system management team investigated, recommended, and maintained state-of-the-art equipment such as workstations, printers, disk drives, memory, graphics cards, monitors, CPU upgrades, keyboards, ethernet cards, and tape back-up systems. The team managed network services including printing, user accounting, server management, hardware and software installation, backups, and data recovery. Software packages, such as FLUENT, FIDAP, POLYFLOW, and STAR-CD, were maintained. Additionally, the team investigated and evaluated new software and hardware as these became available, making recommendations as to which should be acquired to enhance the capabilities of the CML and the AMRL at The University of Akron.
Bibliography Type:	Description: (Last Updated: 04/03/2007)
Articles in Peer-reviewed Journals	Buldum,A.; Busuladzic,I.;Clemons,C.B.; Dill,L.H.; Kreider,K.L.; Young,G.W.; Evans,E.A.; Zhang,G.; Hariharan,S.I.; Keifer,W. "Multi-Scale Modeling, Simulations and Experiments of Coating Growth on Nanofibers: Part I - Sputtering" J. Applied Physics, Vol. 98 (2005), pp. 044303-044303-10 , Aug-2005
Articles in Peer-reviewed Journals	Buldum,A.; Clemons,C.B.; Dill,L.H.; Kreider,K.L.; Young,G.W., X. Zheng;Evans,E.A.; Zhang,G.; Hariharan,S.I. "Multi-Scale Modeling, Simulations and Experiments of Coating Growth on Nanofibers: Part II - Deposition" J. Applied Physics, Vol. 98 (2005), pp. 044304-044304-16 , Aug-2005
Articles in Peer-reviewed Journals	Marinov,T.; Buldum,A.;Clemons,C.B.; Kreider,K.L.; Young,G.W.; Hariharan,S.I. "Field Emission from Coated Nanowires" J. Applied Physics, Vol. 98 (2005), pp. 044314-044314-11 , Aug-2005